

Reasoning Reference: Analogies

Educational Series

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1 Introduction

An **Analogy** literally means 'similarity'. In reasoning, an analogy question presents a specific relationship between two given items and requires you to find a similar relationship between another pair. It is typically represented as:

$$A : B :: C : D$$

(Read as: A is to B as C is to D)

2 Common Types of Word Analogies

2.1 Synonyms and Antonyms

- **Synonyms:** Abbreviate : Shorten :: Precise : Exact
- **Antonyms:** Kindle : Extinguish :: Chaos : Order

2.2 Worker and Tool/Product

- **Worker and Tool:** Sculptor : Chisel :: Carpenter : Saw
- **Worker and Product:** Architect : Design :: Chef : Meal

2.3 Quantity and Unit

- **Metric:** Current : Ampere :: Power : Watt
- **Metric:** Resistance : Ohm :: Force : Newton

2.4 Instrument and Measurement

- **Device:** Barometer : Pressure :: Thermometer : Temperature
- **Device:** Anemometer : Wind Speed :: Hygrometer : Humidity

2.5 Animal and Young One/Group/Sound

- **Young Ones:** Cat : Kitten :: Frog : Tadpole
- **Sounds:** Lion : Roar :: Horse : Neigh

3 Number Analogies

Number analogies involve mathematical relationships like squares, cubes, prime numbers, or arithmetic operations.

- **Example (Squares):** $25 : 625 :: 35 : ?$
Pattern: $x : x^2 \implies 35^2 = 1225$.
- **Example (Operations):** $6 : 18 :: 4 : ?$
Pattern: $(x^2 \div 2) \implies (4^2 \div 2) = 8$.

4 Letter Analogies

These follow the same logic as *Coding and Decoding*, relying on alphabet positions and shifting.

- **Example:** BC : FG :: HI : ?
Pattern: B(+4)=F, C(+4)=G. Apply to H(+4)=L, I(+4)=M.
Answer: LM

5 Solving Strategy

1. **Define the Relationship:** Create a sentence linking the first two words (e.g., "A *chisel* is a tool used by a *sculptor*").
2. **Check Order:** Ensure the relationship flows in the same direction (Worker:Tool must match Worker:Tool, not Tool:Worker).
3. **Eliminate Options:** Rule out choices that have no internal relationship or the wrong type of relationship.
4. **Identify the Best Fit:** If two options seem correct, look for a more specific secondary relationship (e.g., Degree of intensity).

6 Solved Examples

Example 1: Odometer : Distance :: Compass : ?

Logic: An Odometer measures Distance. Similarly, a Compass measures Direction.

Answer: Direction

Example 2: 8 : 28 :: 27 : ?

Logic: $2^3 : (3^3 + 1)$. The relationship is $n^3 : ((n + 1)^3 + 1)$.

Apply to 27 (which is 3^3): $(3 + 1)^3 + 1 = 4^3 + 1 = 64 + 1 = 65$.

Answer: 65